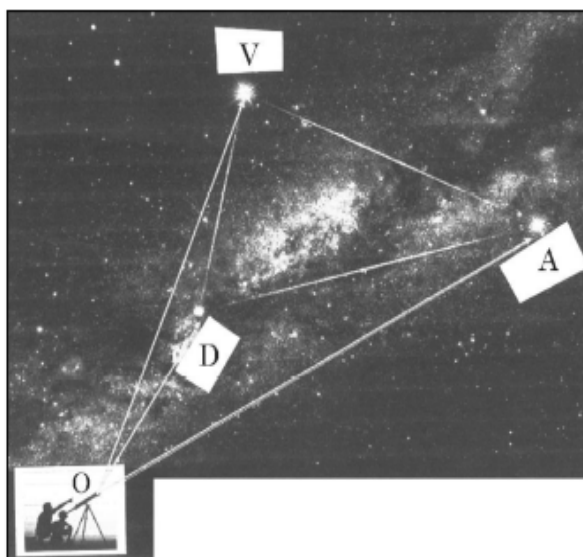


UNIT-4 VECTOR AND 3-D

CASE BASED QUESTIONS

1. An instructor at the astronomical centre shows three among the brightest stars in a particular constellation. Assume that the telescope is located at $O(0, 0, 0)$ and the three stars have their locations at the points D, A and V having position vectors $2\hat{i} + 3\hat{j} + 4\hat{k}$, $7\hat{i} + 5\hat{j} + 8\hat{k}$ and $-3\hat{i} + 7\hat{j} + 11\hat{k}$ respectively.



Based on the above information, answer the following questions :

- (i) How far is the star V from star A ?
- (ii) Find a unit vector in the direction of $\overrightarrow{DA}$.
- (iii) Find the measure of $\angle VDA$.

OR

- (iii) What is the projection of vector $\overrightarrow{DV}$ on vector $\overrightarrow{DA}$?

2. An engineer is designing a new metro rail network in a city.



Initially, two metro lines, Line A and Line B, each consisting of multiple stations are designed. The track for Line A is represented by

$l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{4}$, while the track for Line B is represented by

$l_2 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z+2}{-3}$.

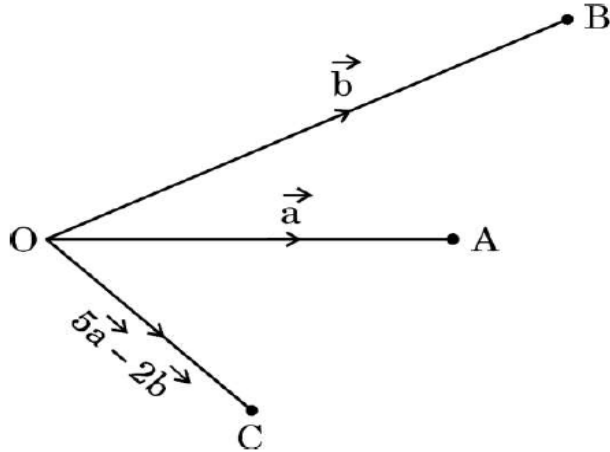
Based on the above information, answer the following questions :

- (i) Find whether the two metro tracks are parallel. 1
- (ii) Solar panels are to be installed on the rooftop of the metro stations. Determine the equation of the line representing the placement of solar panels on the rooftop of Line A's stations, given that panels are to be positioned parallel to Line A's track (l_1) and pass through the point $(1, -2, -3)$. 1
- (iii) (a) To connect the stations, a pedestrian pathway perpendicular to the two metro lines is to be constructed which passes through point $(3, 2, 1)$. Determine the equation of the pedestrian walkway. 2

OR

- (iii) (b) Find the shortest distance between Line A and Line B. 2

3. Three friends A, B and C move out from the same location O at the same time in three different directions to reach their destinations. They move out on straight paths and decide that A and B after reaching their destinations will meet up with C at his predecided destination, following straight paths from A to C and B to C in such a way that $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and $\vec{OC} = 5\vec{a} - 2\vec{b}$ respectively.



Based upon the above information, answer the following questions :

- (i) Complete the given figure to explain their entire movement plan along the respective vectors. 1
 - (ii) Find vectors $\vec{AC}$ and $\vec{BC}$. 1
 - (iii) (a) If $\vec{a} \cdot \vec{b} = 1$, distance of O to A is 1 km and that from O to B is 2 km, then find the angle between $\vec{OA}$ and $\vec{OB}$. Also, find $|\vec{a} \times \vec{b}|$. 2
- OR**
- (iii) (b) If $\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$, then find a unit vector perpendicular to $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$. 2

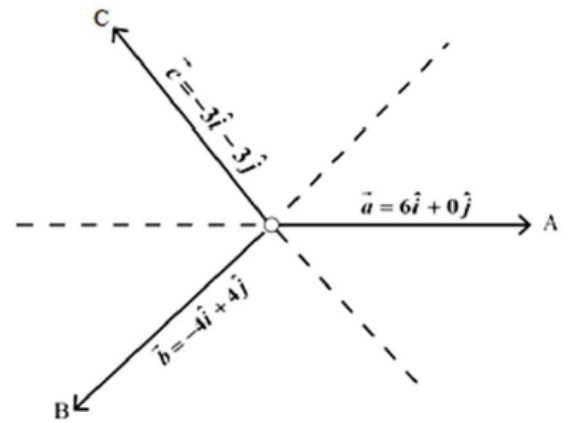
4. Read the following passage and answer the questions given below:

Teams A, B, C went for playing a tug of war game. Teams A, B, C have attached a rope to a metal ring and is trying to pull the ring into their own area.

Team A pulls with force $F_1 = 6\hat{i} + 0\hat{j} \text{ kN}$,

Team B pulls with force $F_2 = -4\hat{i} + 4\hat{j} \text{ kN}$,

Team C pulls with force $F_3 = -3\hat{i} - 3\hat{j} \text{ kN}$,



- (i) What is the magnitude of the force of Team A ?
- (ii) Which team will win the game?
- (iii) Find the magnitude of the resultant force exerted by the teams.

OR

- (iii) In what direction is the ring getting pulled?